OMAR HAMMOUD

omar.hammoudx@gmail.com
+41 78 206 30 53 - Zurich, Switzerland
https://omar-hammoud.vercel.app

EDUCATION

Imperial College London (ICL) *London, United Kingdom*
ETH Master Thesis – Agentic AI in Big Data Processing *Sep. 2026 - Mar. 2027*

Swiss Federal Institute of Technology (ETH Zürich) *Zurich, Switzerland*
Master of Science in Computer Science – Data Management, Security *Sep. 2024 - Mar. 2027*

Swiss Federal Institute of Technology Lausanne (EPFL) *Lausanne, Switzerland*
Bachelor of Science in Computer Science *Sep. 2021 - Aug. 2024*

• **Relevant Courses:** Big Data, Data Management Systems, Cloud Computing Architecture, Software Engineering, Network Security, System Security, High Performance Computing

EXPERIENCE

• **Google - Site Reliability Engineer Intern** – [Dublin, Ireland] *May 2026 - Aug. 2026*
Built and deployed an internal full-stack tool (TypeScript, D3.js, Graphviz, Go) visualizing function-call dependencies between input endpoints, helping engineers trace system behavior.

• **Supercell - Research Engineer** – [Helsinki, Finland] *Mar. 2026 - Apr. 2026*
Built AI-powered tools for 3D asset search and world generation at an educational game studio.

• **ETH Zurich - Teaching Assistant** – [Zurich, Switzerland] *Feb. 2025 - Jan. 2026*
Assisted 400 students enrolled in the Databases and Software Engineering courses by leading exercise sessions and holding office hours to help them with course projects.

• **HTech - Web Developer** – [Beirut, Lebanon] *Jul. 2024 - Oct. 2024*
Built and deployed a client's website using React and .NET, from prototype to delivery.

• **EPFL - Teaching Assistant** – [Lausanne, Switzerland] *Sep. 2023 - Jun. 2024*
Assisted 800 students enrolled in the Computer Architecture and Digital Systems courses through office hours and tutorials focused on advanced topics such as pipelining, caching, and digital logic.

PROJECTS

• **RumbleDB:** [Java] Extensively tested the JSONiq Update Facility, measuring system performance under scaling, and implemented support for Apache Iceberg and Hudi beside Delta Lake.

• **JobsOnCloud:** Evaluated static and dynamic schedulers for memcached and PARSEC workloads on Google Cloud, improving batch job completion while meeting tail-latency SLOs.

• **LikingNodes:** [JavaScript, Python] Developed an interactive web app where users can add and like nodes in live sessions, using `D3.js` for visualization and React, FastAPI, and WebSockets for real-time updates; deployed on Vercel, Railway, and Neon.

SKILLS

• **Programming Languages:**
- *Proficient in:* Python, Java, C, JavaScript, TypeScript, Go, Kotlin, C++, SQL
- *Familiar with:* Scala, C#, Assembly, Verilog, HTML, CSS

• **Technologies:** Git, Docker, Kubernetes, CI/CD, Google Cloud, Cloudflare, PySpark, ReactJS, FastAPI, Android Development, JSONiq, CUDA

• **Hobbies:** Skiing, Running, Swimming, Golf, Gym

• **Languages:** Fluent in French, English and Arabic (native). Learning German